

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Ticks in boxes 3, 4 and 5 i.e. $3 \times (1)$	3			3		
	(b)	(i)	I	8 min 20 s		1		1	1	
			II	6 min 40 s		1		1	1	
		(ii)		Arrival time of P waves: 00 : 16 : 00 (1) Time difference: 00 : 02 : 40 ecf (1) [unless P wave arrival time > 18:40:00] Distance to epicentre: $1.6 \pm 0.2 \times 10^3$ km] ecf from Δt (1) [no ecf on $d > 10 \times 10^3$ km] $1:40 \rightarrow d = 1$ $2:00 \rightarrow d = 1.2$ $d = 1.4, 1.6, 1.8$ in tolerance $3:20 \rightarrow d = 2$			3	3	3	
		(iii)		Conversion to 360 s (1) Speed = 8.89 [km/s] (1) [or with correct rounding] c.a.o.		2		2	2	
		(iv)		2 crosses at intersections		1		1		
		(v)		The top intersection point identified (1) no ecf The distance of the epicentre from BC is shorter than the distances of MC and BH [or circle centred on BC crossing through the top intersection point seen] (1) NB. If a 3rd circle is drawn which does not have a common intersection with the others, identification + justification e.g. it is the closer point.			2	2	1	
				Question 8 total	3	5	5	13	8	0